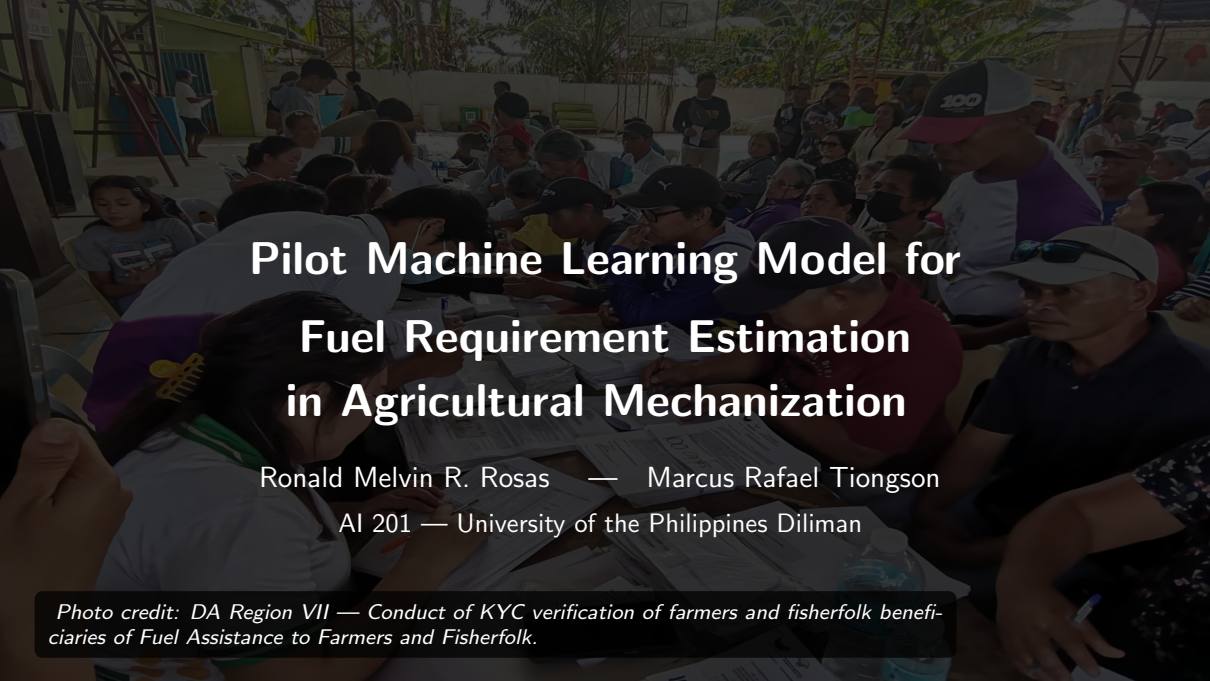




Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

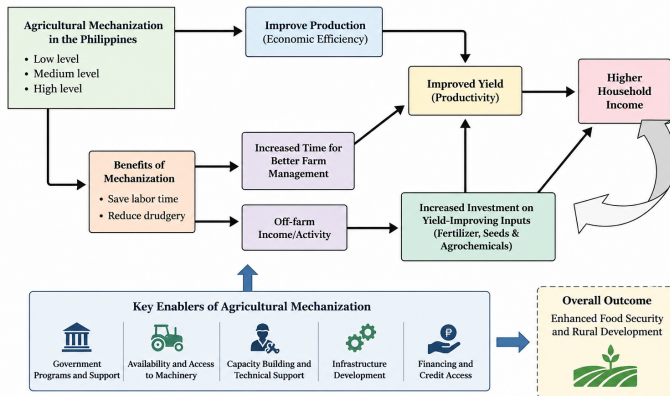
Ronald Melvin R. Rosas — Marcus Rafael Tiongson

AI 201 — University of the Philippines Diliman

Photo credit: DA Region VII — Conduct of KYC verification of farmers and fisherfolk beneficiaries of Fuel Assistance to Farmers and Fisherfolk.

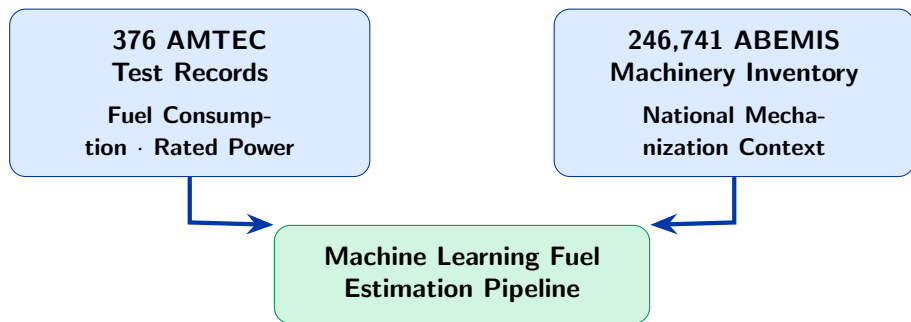
Agricultural Mechanization in the Philippine Context

Philippine Agricultural Mechanization Context



Fuel-dependent agricultural operations require accurate and scalable fuel requirement estimation methods for mechanization planning, government support programs, and inventory-based decision support.

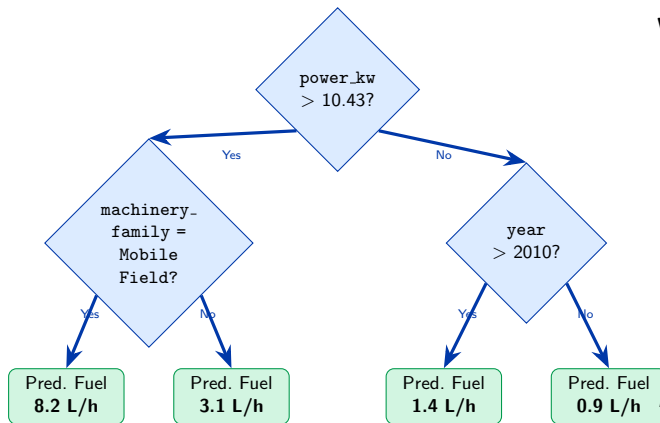
The Data: AMTEC & ABEMIS



AMTEC — Philippine machinery testing and evaluation center, based in UPLB, providing standardized agricultural machinery performance and fuel consumption data.
Source: UPLB-AMTEC (1990s–2026)

ABEMIS — National agricultural engineering information system of BAFE for machinery inventory, infrastructure monitoring, and mechanization decision support.
Coverage: all 18 Philippine regions

The Engine: Capturing Non-Linear Physical Interactions



Why not OLS?

- Power and fuel do *not* scale linearly across machine types
- Each tree partitions the feature space differently
- 300 trees average out variance — robust predictions
- No manual feature crossing required

$$R^2_{\text{RF}} = 0.916 \quad \text{vs.} \quad R^2_{\text{OLS}} = 0.704$$

Random Forest vs. OLS Baseline ($n_{\text{test}} = 76$)

Model	R^2	MAE (L/h)	RMSE (L/h)
OLS Global Linear	0.704	1.882	2.875
OLS Hierarchical Fallback	0.703	1.879	2.879
Random Forest	0.916	0.950	1.534

+30%

better R^2

2×

lower MAE

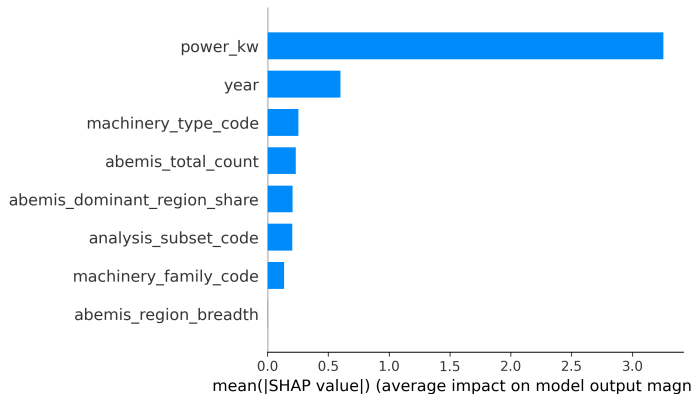
2×

lower RMSE

But why does the model trust these predictions?

Opening the black box with SHAP and LIME.

What Drives the Model? (Validating Physics with SHAP)



Key finding:

power_kw accounts for **~66.5%** of total prediction importance.

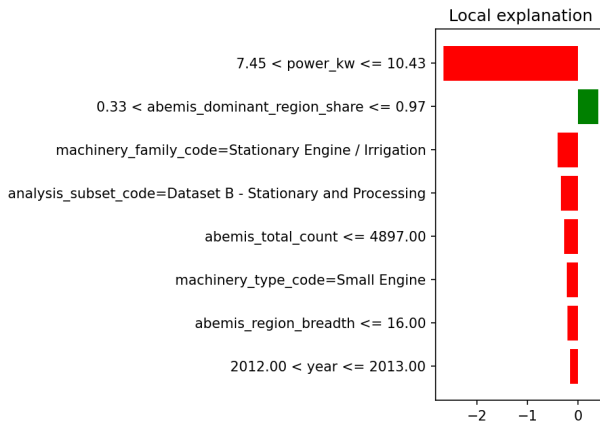
year (equipment vintage) is second at **~12.3%**, reflecting improved combustion efficiency in newer equipment.

ABEMIS context features contribute a combined **~9%**, adding real but secondary value.

The model didn't memorize data — it *learned physics*.

Explaining Individual Predictions (LIME)

LIME - Sample 372 | Small Engine | Actual=5.240 Pred=4.773



LIME breaks down *why* the model predicted **4.77 L/h** for one specific small engine — granular transparency that builds trust for BAfE engineers.

From Lab to National Scale

172,265

machines successfully scored

National inventory: 246,741

Coverage: **70 %**

Missing power data: 62,808

Unmappable type: 11,668

Outputs per region:

- Per-record fuel estimate (L/h)
- Per-region aggregate demand
- Per-region $\times$ machinery-type breakdown

The unscored 30% reflects *data gaps* — not model failure. Closing those gaps is part of the roadmap.

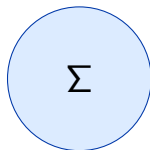
An aerial photograph of a lush mountain valley. In the foreground, a cluster of small, colorful houses with red, blue, and green roofs is nestled among green trees. Behind the village, a series of terraced rice fields, appearing as a patchwork of green and yellow, stretch across the valley floor. The background is dominated by steep, densely forested mountains under a clear sky.

What's next?

Closing the data gaps. Validating in the field.

Future Directions of the Study

The proposed framework may be expanded toward national-scale fuel forecasting, seasonal planning, and data-driven mechanization support.



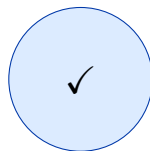
National Fuel Forecasting

Estimate regional and seasonal agricultural fuel demand using national machinery inventories and operational datasets.



Seasonal Planning

Integrate cropping calendars to identify peak planting and harvesting fuel demand periods.



Real-World Validation

Compare machine learning predictions against actual farm-level fuel consumption observations.

Long-term applications may support fuel assistance programs, mechanization planning, and agricultural decision support systems in the Philippines.

Thank you

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